



Pavement Marking Management

Gregory Ciparelli
Connecticut Department of Transportation
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CTDOT/UCONN

A partnership in GIS Data Development & Support

Aaron Nash

aaron.nash@uconn.edu

Dr. Moataz Kilany

moataz.kilany@uconn.edu

Jennifer Pawelzik

jennifer.pawelzik@uconn.edu

Integration with Various Systems

Machine Learning & Computer
Vision Inventory & Segmentation

Spatial Approach to Planning,
Management, and Costing



Considerations for Asset Lifecycle Management

Various Data Inputs Affect Pavement Markings

Maintenance Work

Capital Projects

Pavement Management

Utilize Authoritative LRS as Backbone of Digital Asset

Create Spatially Accurate Representations of Features

Integrate Field Data Specific to Features with Consistent Location Referencing

Provide Centralized Access to Asset Information



Pavement Marking Inventory & Management System

GIS Development MOU

Task 2. Development of a lane marking inventory and data collection/analysis system

Clear, highly visible pavement markings are an integral component of road safety. On August 5, 2022 the Federal Highway Administration published a final rule in the federal register (87 FR47921) announcing minimum maintained retroreflectivity values for pavement markings. The new requirements became effective on September 6, 2022 with a 4 year compliance window. However, states are left to develop their own strategies on how they will become compliant and remain compliant with nighttime minimum retroreflective regulations.

Based on a series of discussions with the CTDOT, **the primary objective of this project is to develop a comprehensive Pavement Marking Index that provides a systematic approach to inventory, assess, and manage the pavement markings on all state roads in Connecticut.** This index will enable CTDOT to efficiently plan and prioritize the maintenance and replacement of pavement markings, thereby enhancing road safety and optimizing resource allocation.



Pavement Marking Inventory & Management System

PRISM

Measure current retro-reflectivity across CT – **Completed**

Digitize all pavement markings in the state ROW – **In progress**

Integrate installation methods into database – **In progress**

Build degradation model to forecast replacement – **In QA**

Design planning application with integrated costing model – **In progress**



Field Data Collection





Current Retroreflectivity

Retro Dashboard - Indicators show count of segments not length

Filter by Retroreflectivity Range
0 - 600

Select a Road
Click to Select

Search...

- ☐ 1 North
- ☐ 1 South
- ☐ 15 North
- ☐ 15 South
- ☐ 175 North
- ☐ 190 East

Reset

Select all

Left: 1-50
69,585

Left: 51-150
355,967

Left: 151-250
134,076

Left: 251-350
66,902

Left: 351+
91,145

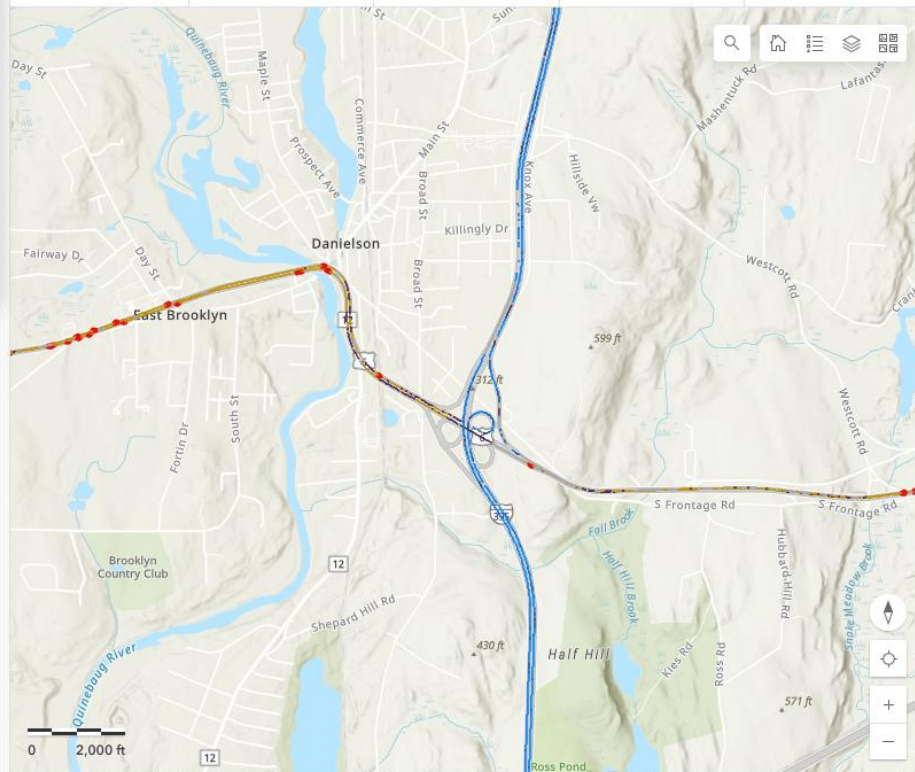
Right: 1-50
42,768

Right: 51-150
282,722

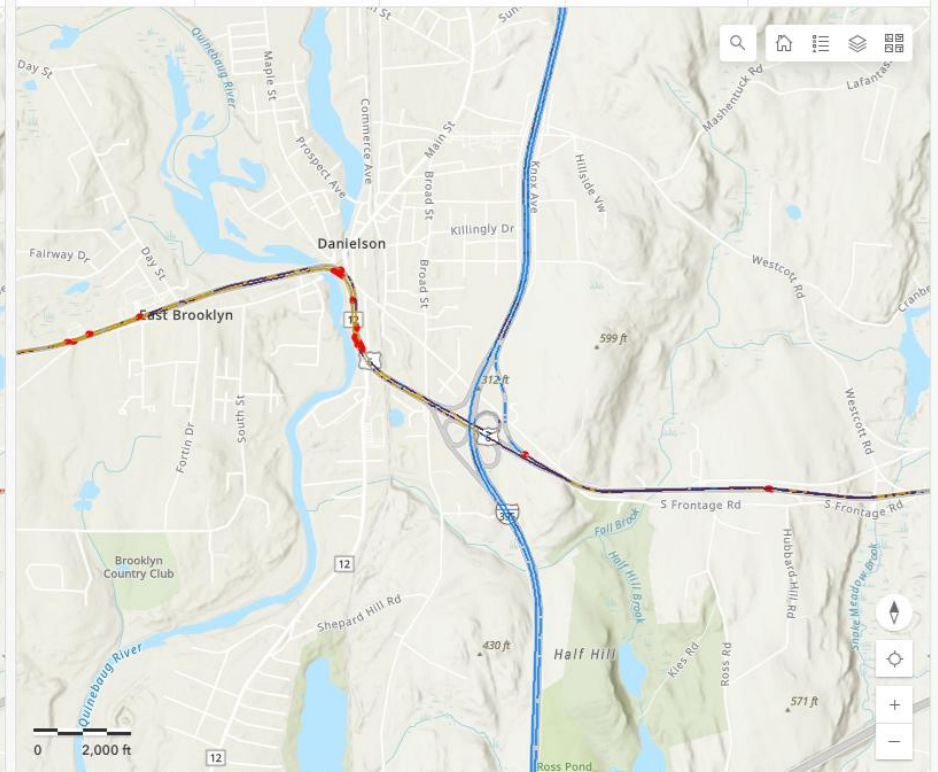
Right: 151-250
149,283

Right: 251-350
90,581

Right: 351+
140,759



Esri, NASA, NGA, USGS, FEMA | Esri, TomTom, Garmin, SafeGraph, GeoTechnologies, Inc, METI/NASA, USGS, EPA, NPS, US C... Powered by Esri



Esri, NASA, NGA, USGS, FEMA | Esri, TomTom, Garmin, SafeGraph, GeoTechnologies, Inc, METI/NASA, USGS, EPA, NPS, US C... Powered by Esri



Digitize Pavement Markings

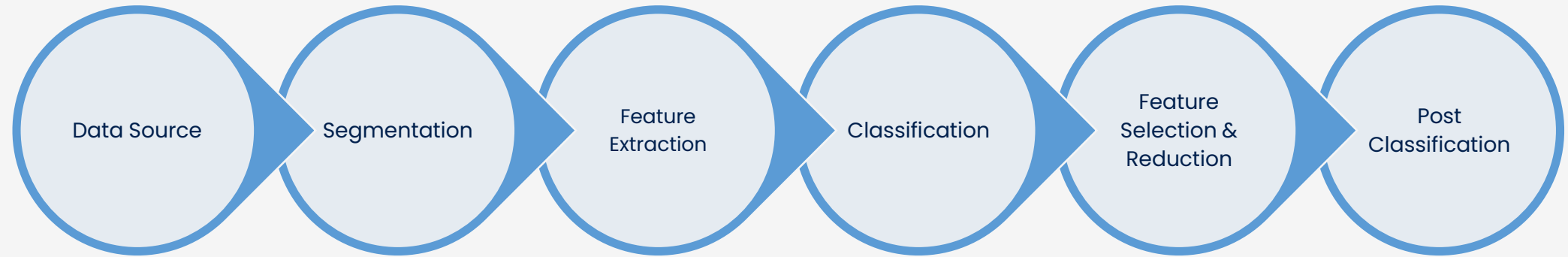


- Stop Line
- Stop Yield
- Solid Yellow
- Arrow One Direction
- Arrow Two Direction
- Double Solid Yellow
- Broken Yellow
- Double Broken Yellow
- Solid White
- Double Solid White
- Broken White
- Asphalt
- Crosswalk
- Guiderail
- Grass
- Shadow
- Dotted White
- Control Box
- Vehicle
- Parking Marks
- Concrete
- Catch Basin

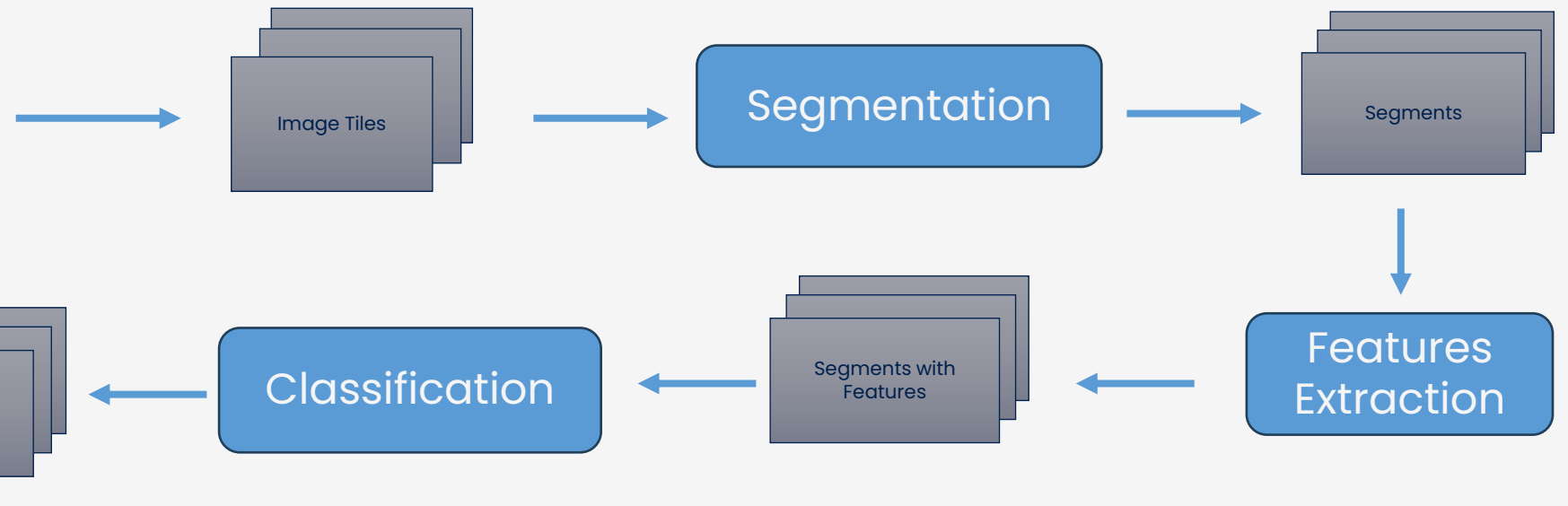




Modeling Overview



Tile1,1	Tile1,2	...1,n
Tile2,1	.	.
Tilem,1	.	.m,n





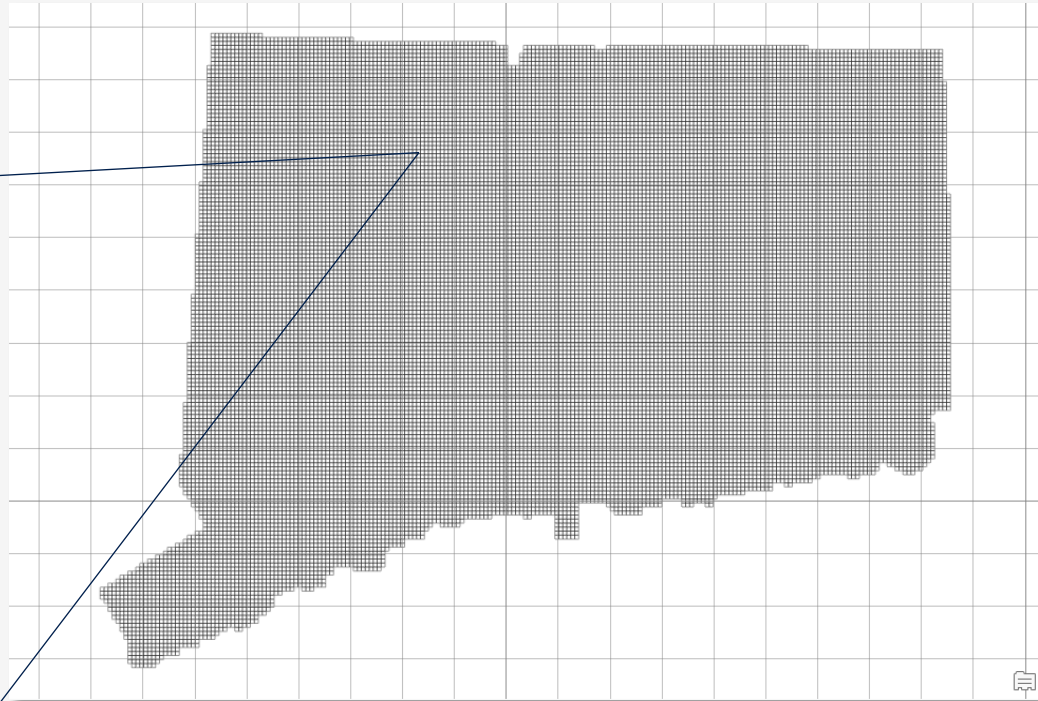
Pavement Marking Modeling – Sources

2023 Aerial Imagery – State of Connecticut – GIS Office.

- 23381 Tiles. 1000x1000 pixel – 2500 x 2500 ft
- Spatial resolution 3 – Inch.
- Spectral resolution 8 – bit.
- Bands : R, G, B, NIR.

LiDAR Imagery.

- 15 Points / Square Meter.



Data Source

Segmentation

Feature Extraction

Classification



Segmentation

Multi-Resolution Image segmentation

- Source Data
 - Digital value of spectral bands.
 - LiDAR
 - Average Intensity



Data Source

Segmentation

Feature
Extraction

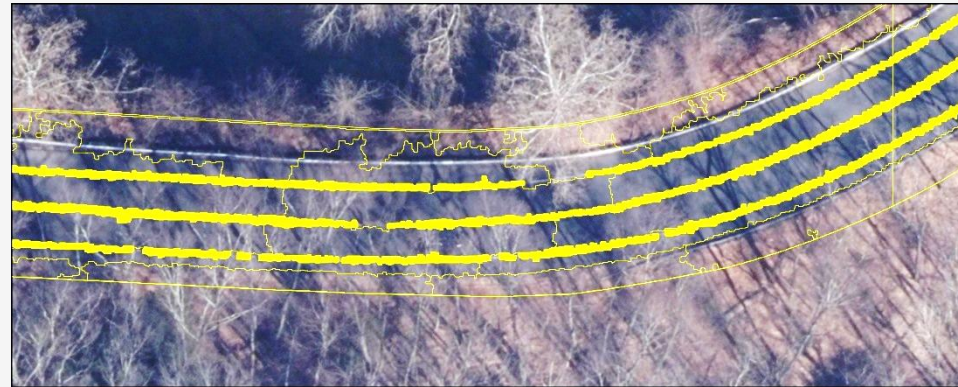
Classification



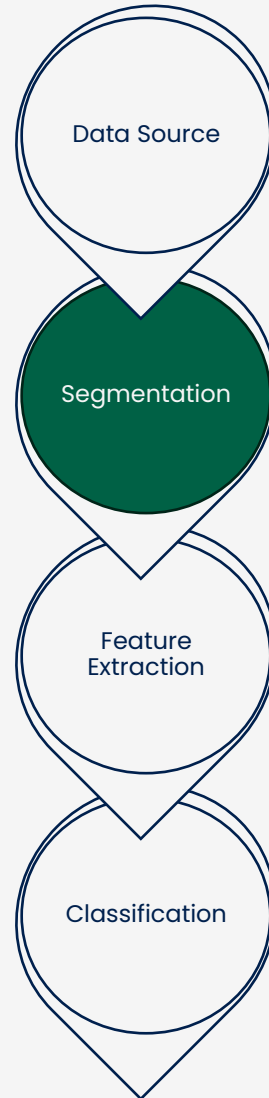
Segmentation – Spectral Imagery vs LiDar



Spectral Based Segmentation



LiDAR Based Segmentation

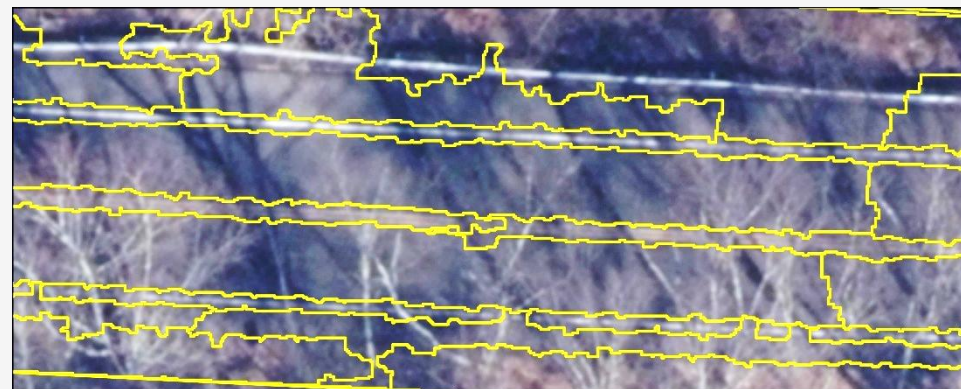




Segmentation – Spectral Imagery vs LiDar



Spectral Based Segmentation



LiDAR Based Segmentation





Extracting Features from Objects

RGB . NIR Spectral Features.

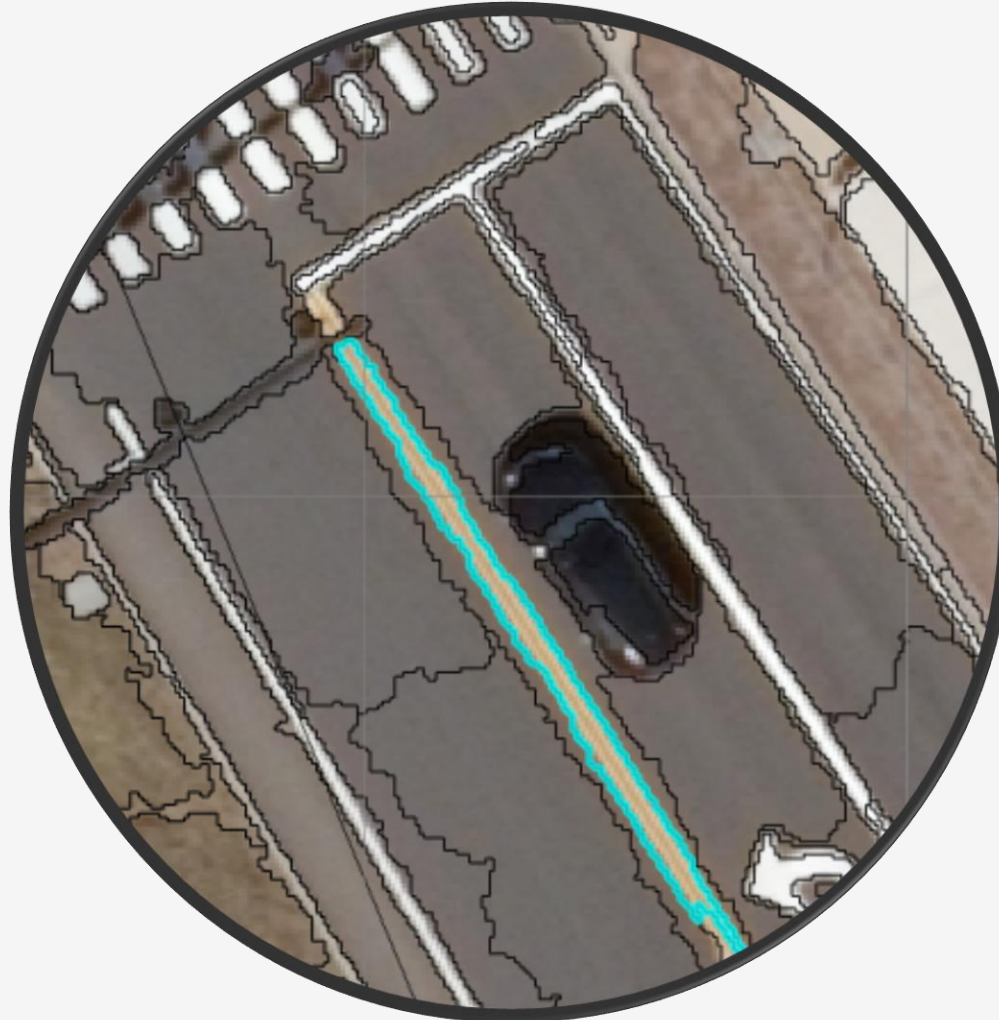
- NDVI (Vegetation Index).
- NDYI (Yellowness Index).
- NDPI (Phenology Index).
- Others.
- Mean, standard deviation, homogeneity and entropy.

Geometric features.

- Area
- Roundness
- Length / Width

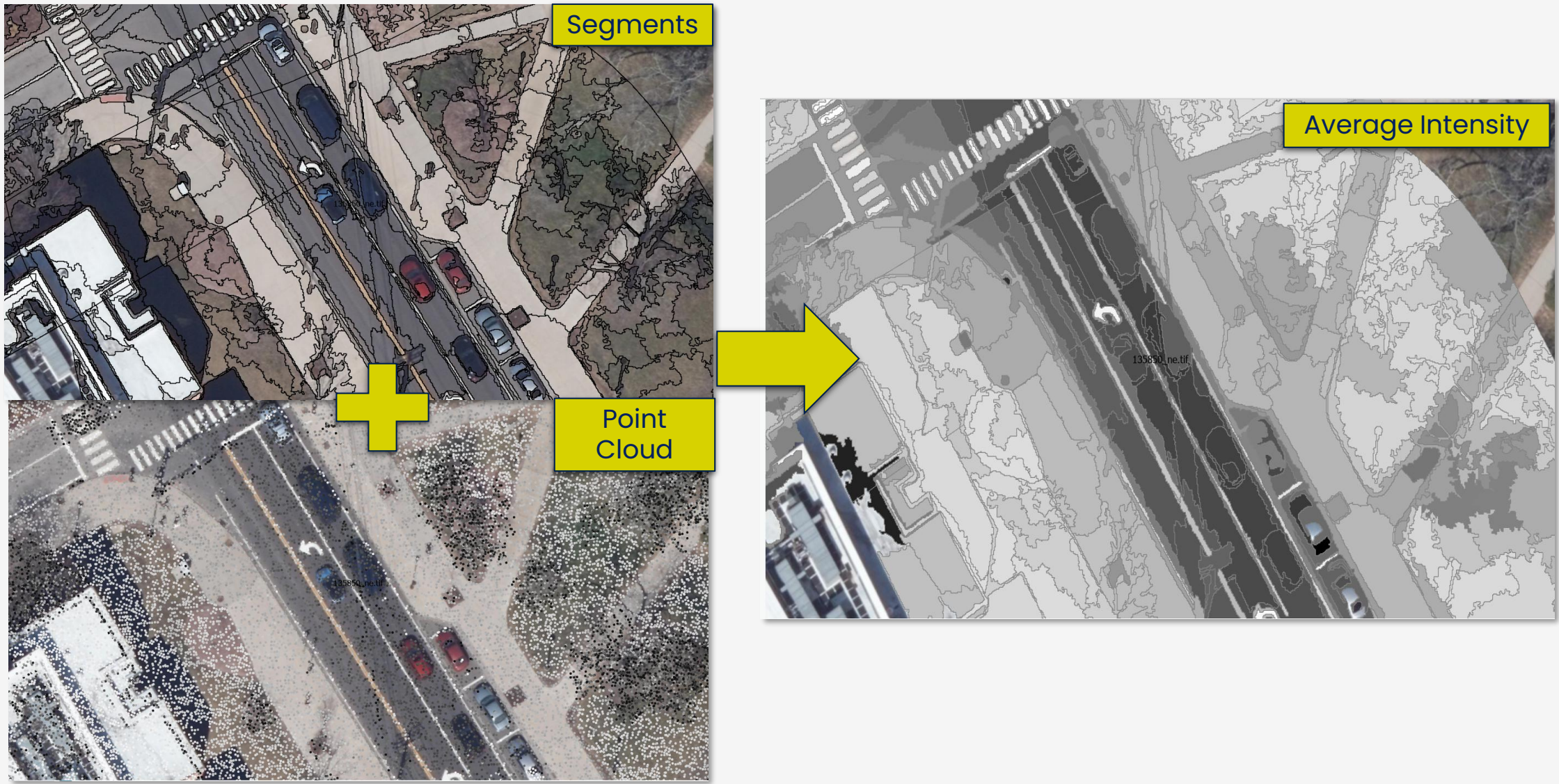
LIDAR

- Mean
- STD
- Min
- Max





LiDar Feature Extraction





Classification



- Stop Line
- Stop Yield
- Solid Yellow
- Arrow One Direction
- Arrow Two Direction
- Double Solid Yellow
- Broken Yellow
- Double Broken Yellow
- Solid White
- Double Solid White
- Broken White
- Asphalt
- Crosswalk
- Guiderail
- Grass
- Shadow
- Dotted White
- Control Box
- Vehicle
- Parking Marks
- Concrete
- Catch Basin

Data Source

Segmentation

Feature
Extraction

Classification



Classification



- Stop Line
- Stop Yield
- Solid Yellow
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Data Source

Segmentation

Feature
Extraction

Classification



Pavement Marking Optimization

Point Intensity Raster Enhancements

- Median + Gaussian (Denoise & Smooth)
- Unsharp Mask (Enhance Edges)
- Morphological Closing (Connect broken markings)
- Max Filter (Strengthen faint connections)
- Gamma Correction (Adjust tone)

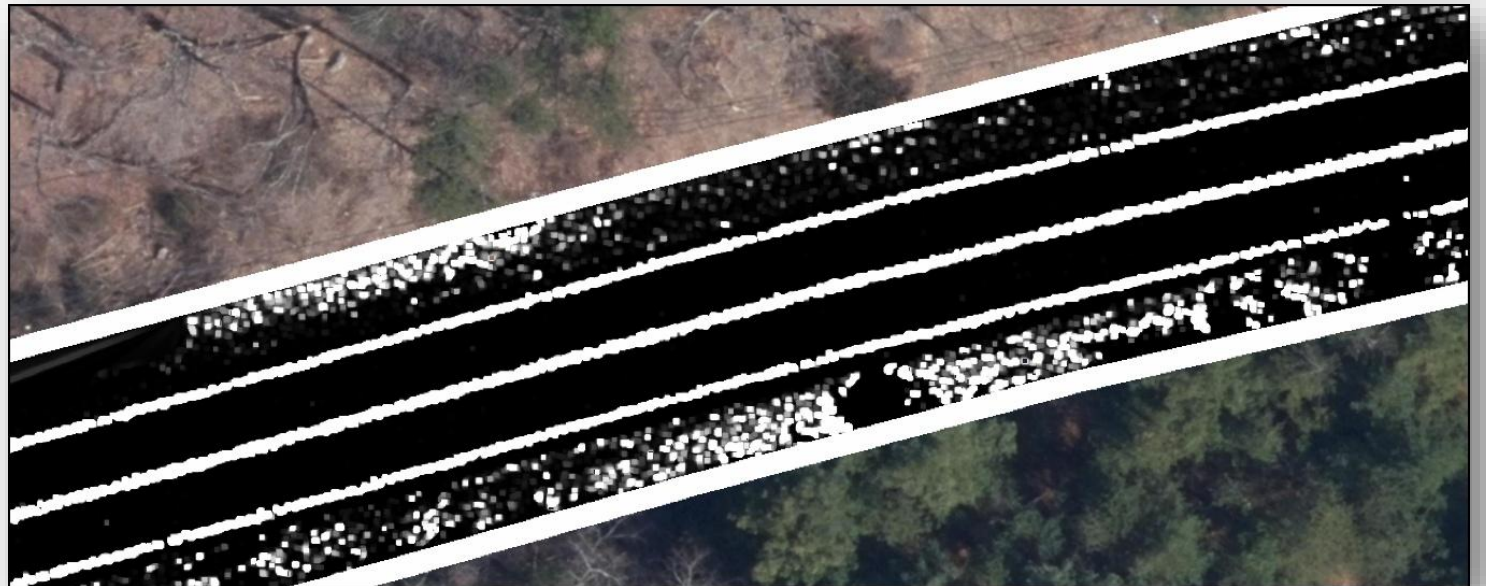
Automated Clean Up Work

Broken whites / Dashed lines

- Auto rotate and adjust length

Enhanced classification

Gap filling





LiDar Limitations

Stop Bars classified as solid white

Severely degraded markings are misclassified

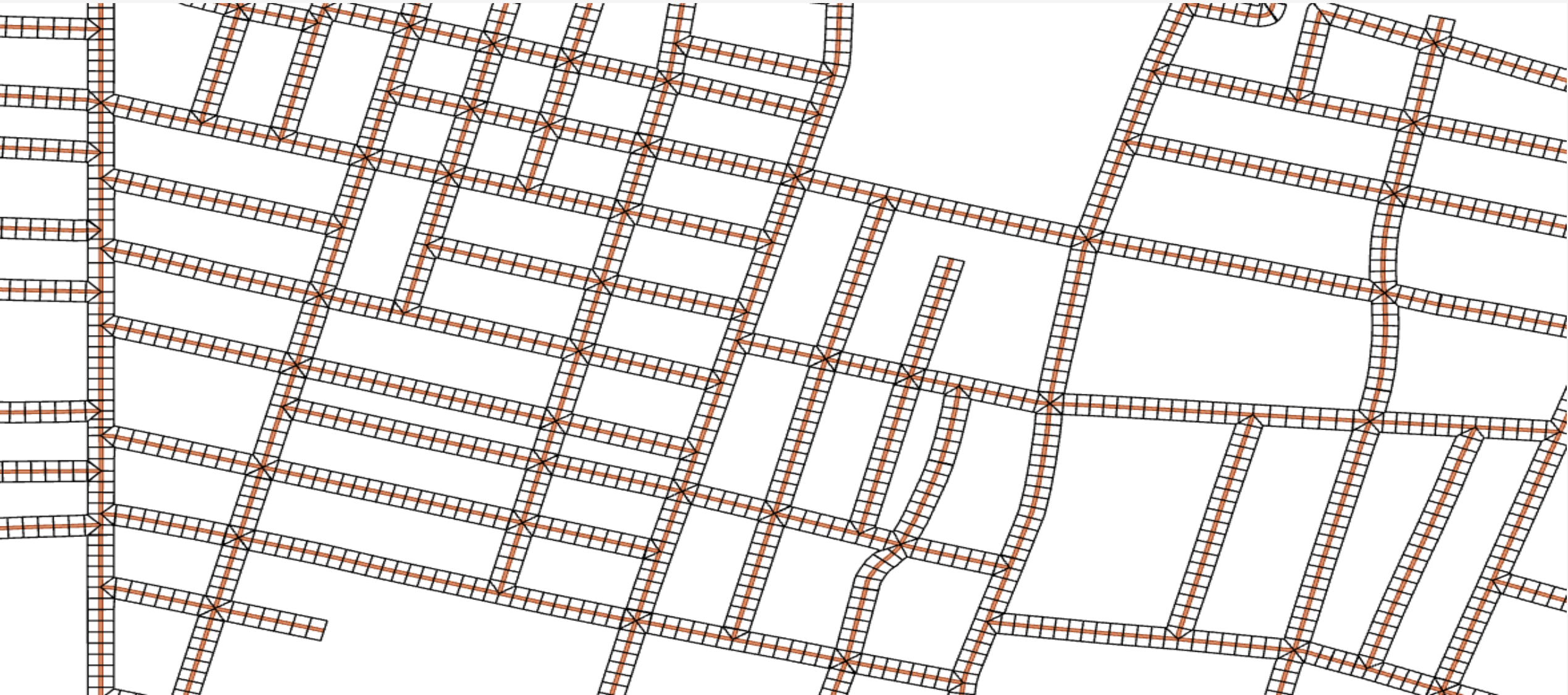
Broken yellow / solid yellow (passing zones) are misclassified as double yellow

All symbols are not collected with Lidar, separate process to accommodate

Spatial and temporal differences between imagery and Lidar raster

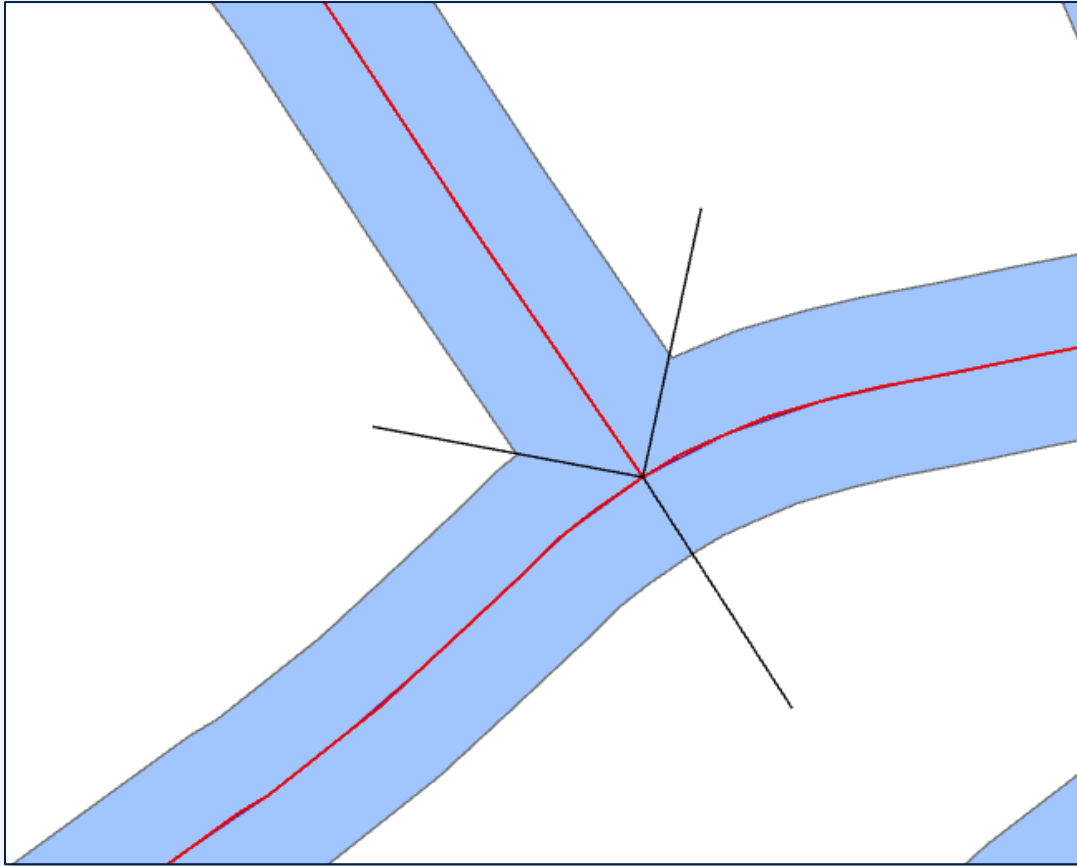


Mile Marker Segmentation





Divide Intersections Consistently



```
import arcpy
import math
import os
from collections import defaultdict

# === Parameters ===
point_fc = arcpy.GetParameterAsText(0)
roads_fc = arcpy.GetParameterAsText(1)
transect_length = float(arcpy.GetParameterAsText(2))
output_transects = arcpy.GetParameterAsText(3)
gdb = arcpy.GetParameterAsText(4)
workspace = gdb
arcpy.env.overwriteOutput = True

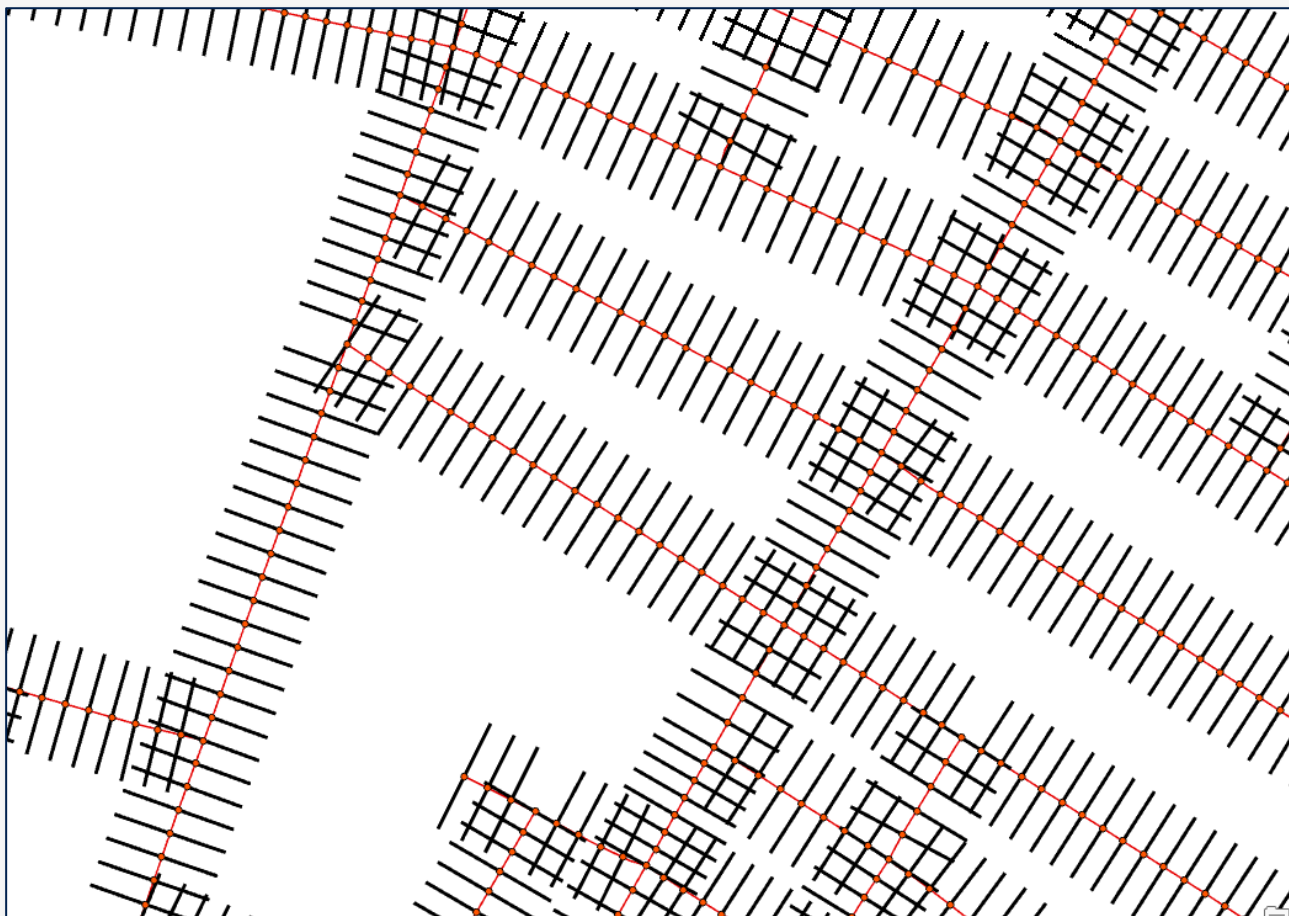
# === Constants ===
angle_tolerance = 2 # degrees
distance_tolerance = 0.1 # meters
search_radius = 5 # meters

# === Helper Functions ===
def unit_vector(dx, dy):
    mag = math.hypot(dx, dy)
    return (dx / mag, dy / mag) if mag != 0 else (0, 0)
def vector_angle(dx, dy):
    return math.atan2(dy, dx) % (2 * math.pi)
def average_vectors(v1, v2):
    x = v1[0] + v2[0]
    y = v1[1] + v2[1]
    return unit_vector(x, y)
def create_transect(origin_point, vector, length):
    dx, dy = vector
    end_point = arcpy.Point(origin_point.X + dx * length, origin_point.Y + dy * length)
    return arcpy.Polyline(arcpy.Array([origin_point, end_point]), 6434)
def is_angle_close(a1, a2, threshold_deg):
    delta = abs(a1 - a2) % 360
    return min(delta, 360 - delta) < threshold_deg

# === Create Output Feature Class ===
spatial_ref = arcpy.Describe(point_fc).spatialReference
mem_fc = arcpy.CreateFeatureclass_management("in_memory", output_transects, "POLYLINE", spatial_reference=spatial_ref).getOutput(0)
arcpy.AddField_management(mem_fc, "PointID", "LONG")
arcpy.AddField_management(mem_fc, "Angle1", "DOUBLE")
arcpy.AddField_management(mem_fc, "Angle2", "DOUBLE")
arcpy.AddField_management(mem_fc, "MidAngle", "DOUBLE")
```



Mile Point Transects



```
# Process points and build transects
count = 0
with arcpy.da.SearchCursor(points_with_near, ["SHAPE@XY", "NEAR_FID", "NEAR_X", "NEAR_Y"]) as search_cursor:
    with arcpy.da.InsertCursor(output_fc, ["SHAPE@"] + field_names) as insert_cursor:

        for row in point_cursor:
            count += 1
            if count % 500 == 0:
                arcpy.AddMessage(f"Processed {count} points...")

            pt_xy, near_fid, near_x, near_y = row[:4]
            attr_vals = list(row[4:])

            # Type conversion for fields
            for i, field in enumerate(field_names):
                expected_type = transfer_fields[field]
                input_type = input_field_types.get(field, None)
                if expected_type == "DOUBLE" and input_type in ("String", "Text"):
                    try:
                        attr_vals[i] = float(attr_vals[i]) if attr_vals[i] is not None else None
                    except:
                        attr_vals[i] = None

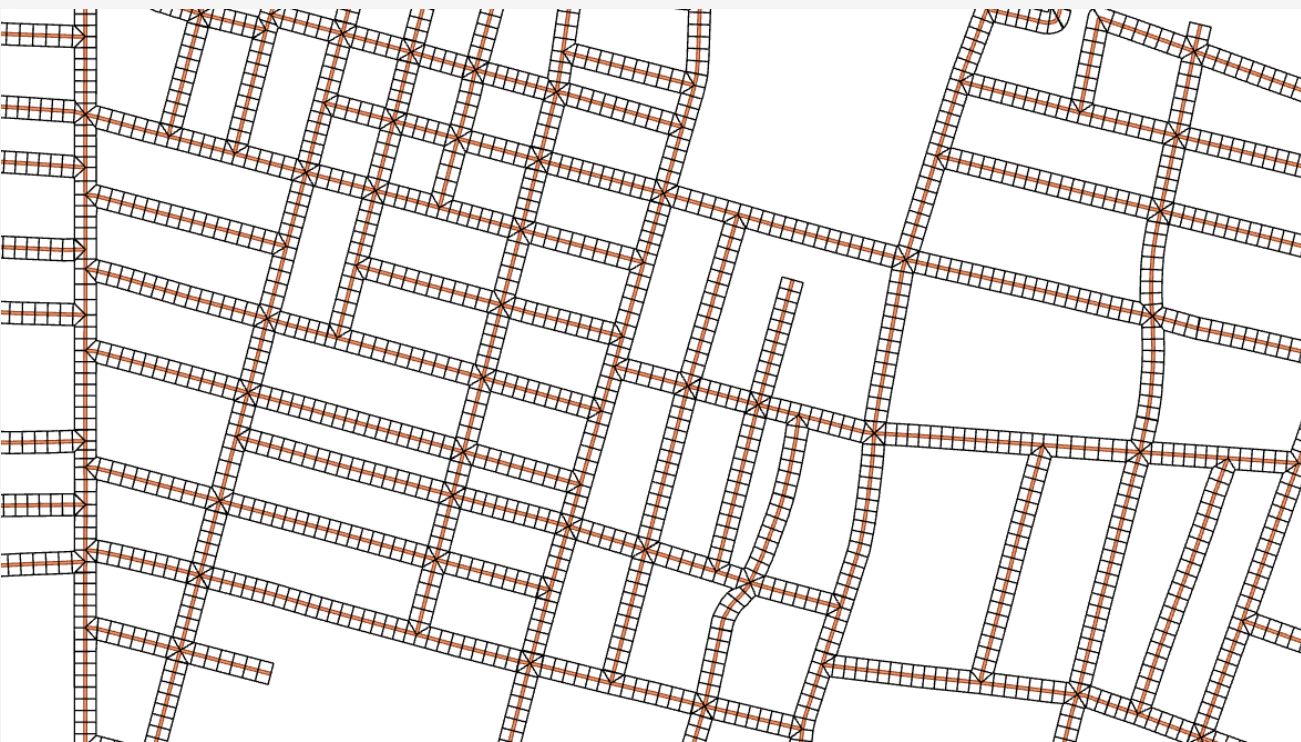
            if near_fid == -1 or near_fid not in line_geom_dict:
                continue

            point_geom = arcpy.PointGeometry(arcpy.Point(*pt_xy), spatial_ref)
            near_point = arcpy.PointGeometry(arcpy.Point(near_x, near_y), spatial_ref)
            line_geom = line_geom_dict[near_fid]

            # Find closest segment on the matched line
            closest_segment = None
            closest_distance = float("inf")
            for part in line_geom:
                for i in range(len(part) - 1):
                    pt1 = part[i]
                    pt2 = part[i + 1]
                    seg = arcpy.Polyline(arcpy.Array([pt1, pt2]), spatial_ref)
                    dist = seg.distanceTo(near_point)
                    if dist < closest_distance:
                        closest_distance = dist
```



Divide Buffered Roadways



```
# Get a list of unique IDs from the polygon dataset
unique_ids = [row[0] for row in arcpy.da.SearchCursor(polygon_fc, route_ID)]
output_fc = str(arcpy.management.CreateFeatureclass("in_memory", output_fcName, "POLYGON", "", has_m, f

# Add fields from both input datasets to the output
polygon_fields = [f.name for f in arcpy.ListFields(polygon_fc) if f.type not in ("Geometry", "OID")]
centerline_fields = [f.name for f in arcpy.ListFields(centerline_fc, field_type=["Double", "String"])

# for field in centerline_fields:
arcpy.AddField_management(output_fc, "ROUTE_ID", "TEXT", field_length=30)
total_ids = len(unique_ids)

# Loop through each unique ID
for idx, unique_id in enumerate(unique_ids):
    percent_complete = (idx + 1) / total_ids * 100
    print(f"Processing ID: {unique_id} ({percent_complete:.2f}% complete)")
    # Filter polygons and transects by the current ID
    route_ID_query = f"{route_ID} = '{unique_id}'"
    # Create in-memory layers for the filtered features
    polygon_layer = "polygon_layer"
    transect_layer = "transect_layer"
    centerline_layer = "centerline_layer"
    arcpy.management.MakeFeatureLayer(polygon_fc, polygon_layer, route_ID_query)
    arcpy.management.MakeFeatureLayer(transect_fc, transect_layer, route_ID_query)
    arcpy.management.MakeFeatureLayer(centerline_fc, centerline_layer, route_ID_query)
    # Temporary output for Feature to Polygon
    temp_output = f"memory\\splitpolygon"
    # Combine polygons and transects to form polygons
    arcpy.FeatureToPolygon_management([polygon_layer, transect_layer], temp_output, "", True)

    # Perform a spatial join to combine attributes from both datasets
    joined_output = f"memory\\joined_polygon"

    # Append the result to the output feature class
    arcpy.Append_management(joined_output, output_fc, "NO_TEST")

arcpy.CopyFeatures_management(output_fc, output_fcPath)
arcpy.AddMessage("Processing complete.")
```



Join Spatially

ROUTE_ID	START_RouteID	END_RouteID	START_MilePoint	END_MilePoint	START_TownName	END_TownName
84-W	84-W	84-W	3.12	3.13	DANBURY	DANBURY
84-W	84-W	84-W	3.11	3.12	DANBURY	DANBURY
84-W	84-W	84-W	3.13	3.14	DANBURY	DANBURY
84-W	84-W	84-W	3.1	3.11	DANBURY	DANBURY
84-W	84-W	84-W	3.09	3.1	DANBURY	DANBURY
84-W	84-W	84-W	3.14	3.15	DANBURY	DANBURY
84-W	84-W	84-W	3.15	3.16	DANBURY	DANBURY
84-W	84-W	84-W	3.08	3.09	DANBURY	DANBURY
84-W	84-W	84-W	3.16	3.17	DANBURY	DANBURY
84-W	84-W	84-W	3.07	3.08	DANBURY	DANBURY
84-W	84-W	84-W	3.17	3.18	DANBURY	DANBURY
84-W	84-W	84-W	3.06	3.07	DANBURY	DANBURY
84-W	84-W	84-W	3.18	3.19	DANBURY	DANBURY
84-W	84-W	84-W	3.05	3.06	DANBURY	DANBURY
84-W	84-W	84-W	3.19	3.2	DANBURY	DANBURY
84-W	84-W	84-W	3.04	3.05	DANBURY	DANBURY
84-W	84-W	84-W	3.2	3.21	DANBURY	DANBURY



Conflate Retroreflectivity Values

...	🔒 End Mile Point	⬇️ ...	🔒 Start Town	⬇️ ...	🔒 End Town	⬇️ ...	🔒 Left Average	⬇️ ...	🔒 Right Average	⬇️ ...	🔒 Left Left	⬇️ ...	🔒 Left Right	⬇️ ...
	0.01		WINDHAM		WINDHAM		37		192		37		0	
	0.02		WINDHAM		WINDHAM		57.2		100.4		57.2		0	
	0.03		WINDHAM		WINDHAM		81.7		29.1		78.6		47.4	
	0.04		WINDHAM		WINDHAM		99.9		9.6		99.6		87.6	
	0.05		WINDHAM		WINDHAM		132.9		23.4		137.9		85.2	
	0.06		WINDHAM		WINDHAM		121.3		53.8		133.3		120.8	
	0.07		WINDHAM		WINDHAM		124.2		31.9		121.8		126.8	
	0.08		WINDHAM		WINDHAM		173.3		48		168.4		178.6	
	0.09		WINDHAM		WINDHAM		206.1		73.9		203.3		130.3	
	0.1		WINDHAM		WINDHAM		120.7		44.2		121.3		15.7	
	0.11		WINDHAM		WINDHAM		154.2		98.2		153.6		43.6	
	0.12		WINDHAM		WINDHAM		208.3		10.8		211.7		147.5	
	0.13		WINDHAM		WINDHAM		57		0		58.2		15.5	
	0.14		WINDHAM		WINDHAM		29.8		0		29.8		0	
	0.15		WINDHAM		WINDHAM		224.8		15.4		234.4		193.3	
	0.16		WINDHAM		WINDHAM		202.5		61.5		205		198.3	
	0.17		WINDHAM		WINDHAM		181.1		0		182.6		123.1	
	0.18		WINDHAM		WINDHAM		191.6		0		185.9		119.9	



Retroreflectivity Data in Segmentation Model

Retroreflectivity Viewer

Left Reflectivity Average Across Connecticut

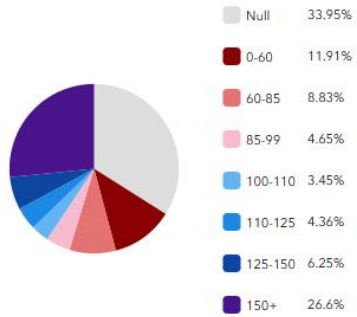


Chart only reflects reflectivity values across entire state. Chart does not change with selection.

Right Reflectivity Average Across Connecticut

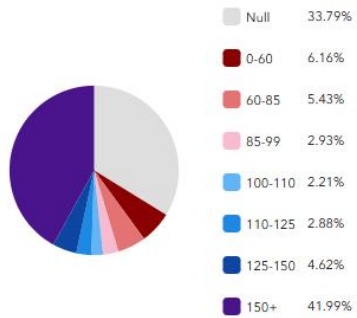


Chart only reflects reflectivity values across entire state. Chart does not change with selection.

Left Millicandela Average in Miles

0-60	60-85	85-99	100-110	110-125	125-150	150+
596.2	459.8	251	172.9	234.7	327.7	134.3k



Right Millicandela Average in Miles

0-60	60-85	85-99	100-110	110-125	125-150	150+
310.0	284.1	158.4	111.1	155.5	242.7	2.1k





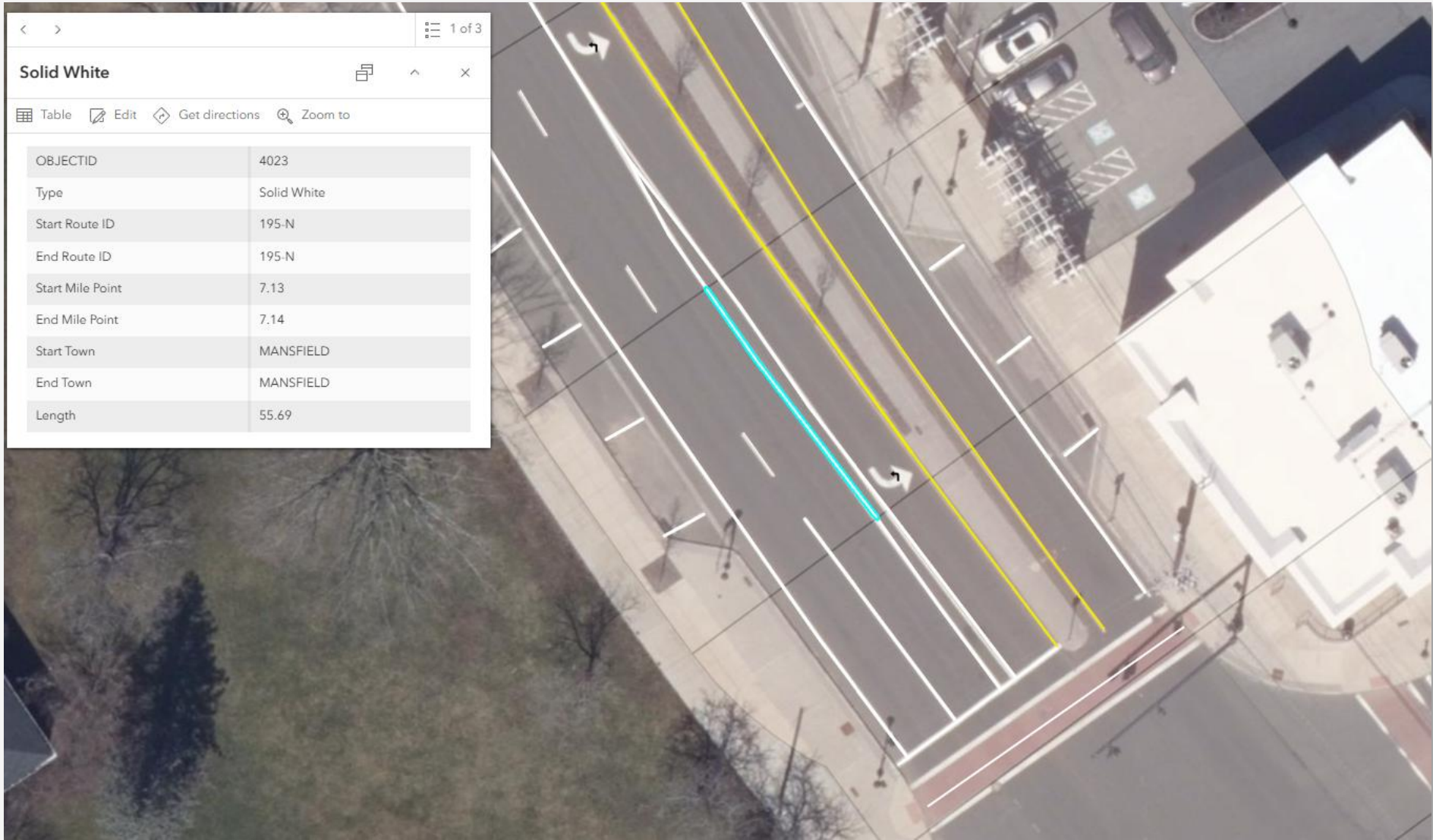
Spatially Accurate Segmented Pavement Markings

< > 1 of 3

Solid White

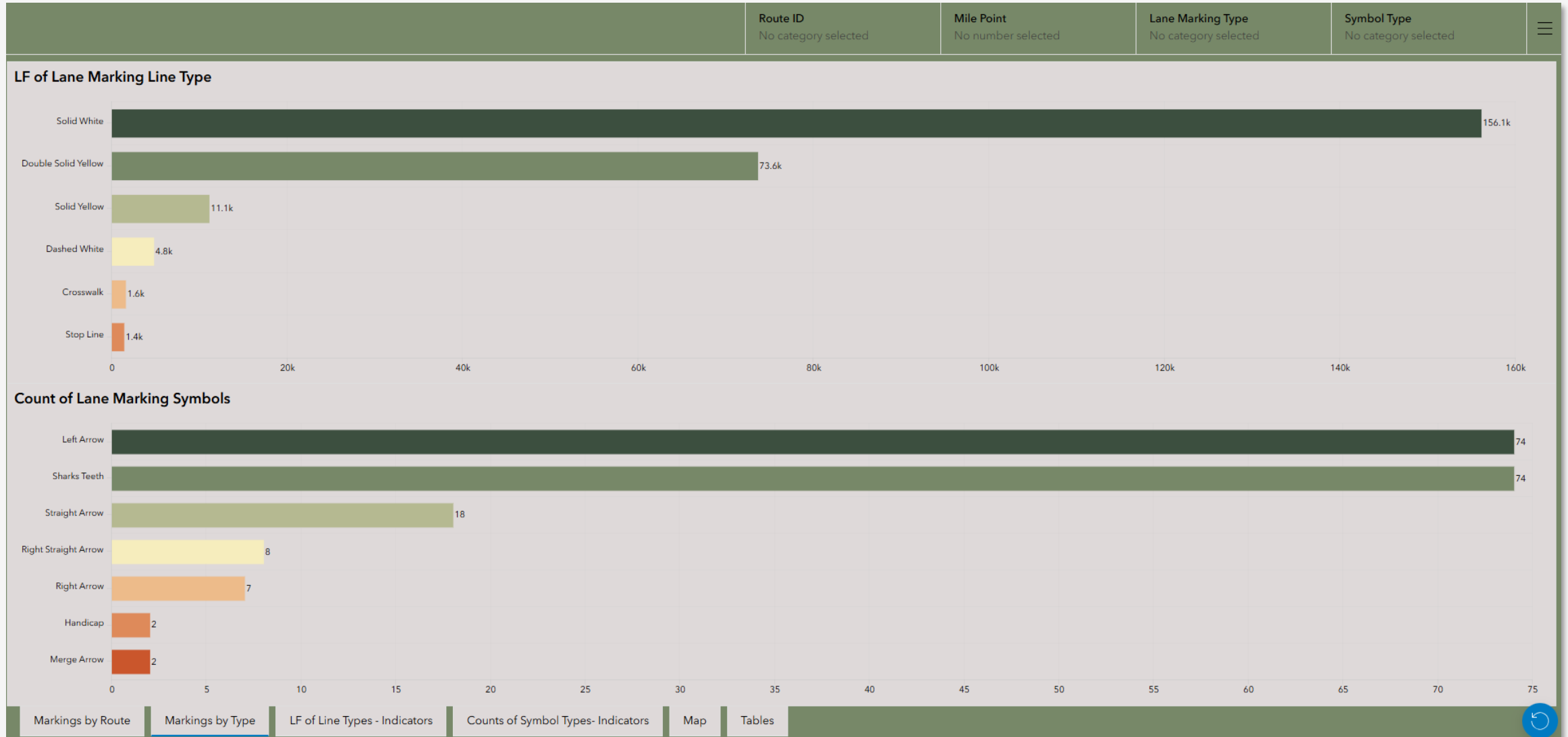
Table Edit Get directions Zoom to

OBJECTID	4023
Type	Solid White
Start Route ID	195-N
End Route ID	195-N
Start Mile Point	7.13
End Mile Point	7.14
Start Town	MANSFIELD
End Town	MANSFIELD
Length	55.69





Quantify By Route & Milepoint





Analysis & Costing Model – Project Identification

Pavement Marking Costing - Contract Number 23PSX0032

Segmentation Model

Planning

Costing

Left Average

Route ID: 1-N
Milepoint Range: 58.12 - 58.15
Left Average:

Route ID: 1-N
Milepoint Range: 70.72 - 71.14
Left Average:

Route ID: 1-N
Milepoint Range: 58.15 - 58.23
Left Average:

Route ID: 1-N
Milepoint Range: 71.14 - 71.15
Left Average:

Route ID: 1-N
Milepoint Range: 58.23 - 58.24
Left Average:

Route ID: 1-N
Milepoint Range: 71.15 - 71.16
Left Average:

Route ID: 1-N
Milepoint Range: 58.24 - 58.33
Left Average:

Route ID: 1-N
Milepoint Range: 71.16 - 71.38
Left Average:

Route ID: 1-N
Milepoint Range: 58.33 - 58.34
Left Average:

Route ID: 1-N
Milepoint Range: 71.38 - 71.39
Left Average:

Route ID: 1-N
Milepoint Range: 58.35 - 58.36
Left Average:



Route ID*

Select a Route number from the dropdown

-Please select-

Starting Mile Point*

Select a Route number from the dropdown

-Please select-

Ending Mile Point*

Select a Route number from the dropdown

-Please select-

Start Location

Find address or place



Aaron



Project Planning & Location

District
No category selected

Project Number
No category selected

Job Type
No category selected

Route ID
No category selected

Milepoints
No number selected

Road Description
No category selected

Paint Type
No category selected

District

2 71%

1 29%

Select a project from the filter list

Select a project from the filter list

Powered by Esri

Projects - 7

District 1 - Project Number: 1
Route ID: 195-N Mile Points 0.02 - 3.54

District 2 - Project Number: 2
Route ID: 195-N Mile Points 10.00 - 15.00

District 2 - Project Number: 2
Route ID: 195-N Mile Points 11.00 - 15.00

District 2 - Project Number: 2
Route ID: 195-N Mile Points 11.00 - 15.00

District 1 - Project Number: 3
Route ID: 195-N Mile Points 4.00 - 6.20

District 2 - Project Number: 4
Route ID: 100-N Mile Points 0.50 - 4.40

District 2 - Project Number: 5
Route ID: 84-E Mile Points 0.28 - 0.91

Project Estimates

Project Number	Marking Type	Line Description	Measurement	Cost per UOM	Estimate
1	Polyurea	Removal of Pavement Markings	550.00 lf	0.45	247.50
1	Polyurea	Removal of Pavement Markings	81.00 sf	1.49	120.69
1	Polyurea	Contractor Furnished Traffic Con...	550.00 lf	0.05	27.50
1	Polyurea	Contractor Furnished Traffic Con...	81.00 sf	0.05	4.05
1	Polyurea	Removal of Pavement Markings	550.00 lf	0	0.00
1	Polyurea	Removal of Pavement Markings	81.00 sf	0	0.00



Multiple Inputs – Centralized Access

Road Name	From	To	Town(s)	Length	LaneMiles	Dist	ADT	PType PMS	2025 PCI	Last Completed Treatment Year	Upcoming Treatment Year (Construction)	Upcoming Program	Upcoming Treatment
084 L	0	0.79	DANBURY	0.79	2.13	4	81,600	COMP	8.7	2025	2030	PRRP	CPR/S
084 L	0.79	3.63	DANBURY	2.84	7.96	4	81,600	COMP	8.7	2025	2030	PRRP	CPR/S
084 L	3.63	8.09	DANBURY	4.46	13.65	4	120,000	COMP	6.8	2015	2029	MRP	MO
084 L	8.09	15.27	DANBURY, BETHEL, BROOKFIELD, NEWTOWN	7.18	17.01	4	76,400	COMP	7.6	2018	2027	PPP	UTB
084 L	15.27	18.04	NEWTOWN	2.77	6.94	4	73,900	COMP	6.8	2018			
084 L	18.04	18.4	NEWTOWN	0.36	0.72	4	73,900	FLEX	7.0	2015			
084 L	18.4	26.49	NEWTOWN, SOUTHBURY, MIDDLEBURY	8.09	21.02	4	69,700	FLEX	5.8	2012	2026	PRRP	CPR/S
084 L	26.49	32.02	MIDDLEBURY, WATERBURY	5.53	13.50	4	71,100	FLEX	8.1	2022	2029	PPP	UTB
084 L	32.02	32.76	WATERBURY	0.74	2.22	4	124,000	COMP	6.8	2015			
084 L	32.76	33.71	WATERBURY	0.95	3.06	4	85,000	COMP	7.0	2019	2026	MRP	MO
084 L	33.71	36.86	WATERBURY	3.15	10.78	4	86,700	COMP	7.0	2016	2026	PPP	UTB
084 L	33.71	40.01	WATERBURY, CHESHIRE	6.30	21.57	4	-	COMP	7.0	2016	2026	PPP	UTB
084 L	36.86	40.01	WATERBURY, CHESHIRE	3.15	10.78	4	100,100	COMP					
084 L	40.01	42.5	CHESHIRE, SOUTHLINGTON	2.49	8.21	1	83,800	COMP					

Pavement Management

Paving treatments affect markings

Forecasted treatments

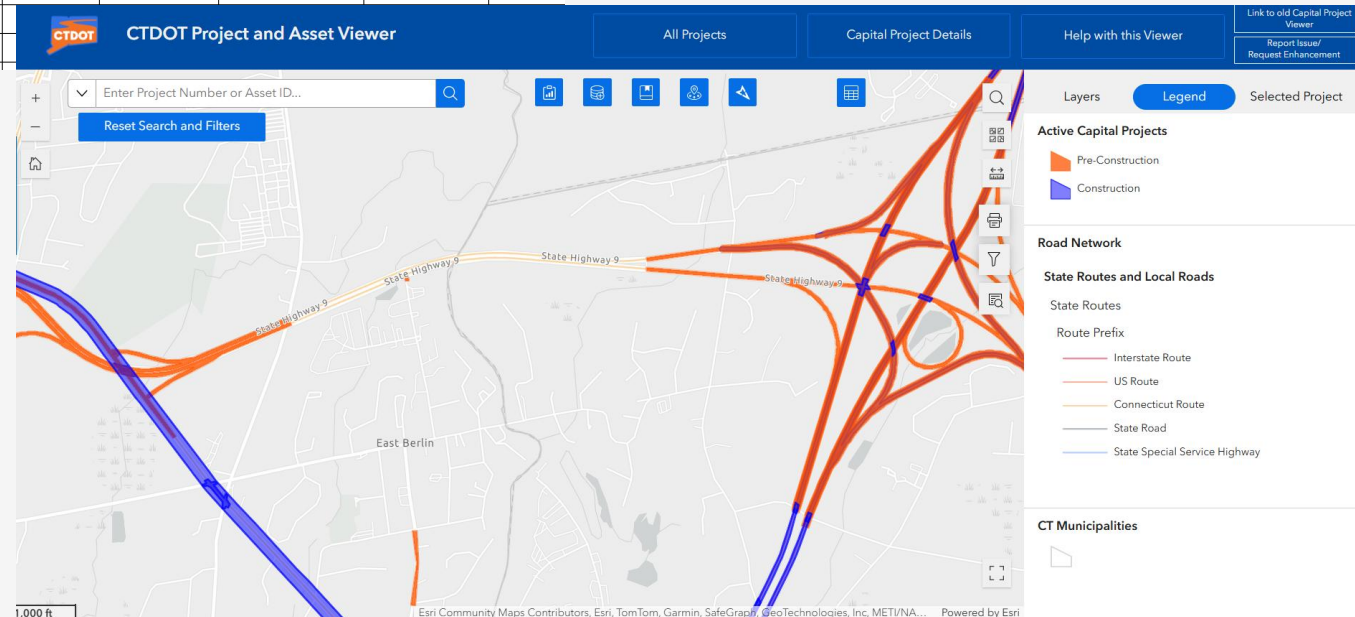
Pavement condition

Capital Project Locations

Planned well into the future

Affect current decision making

Conflict detection





Pavement Markings Degradation Modeling

Forecast out pavement marking degradation

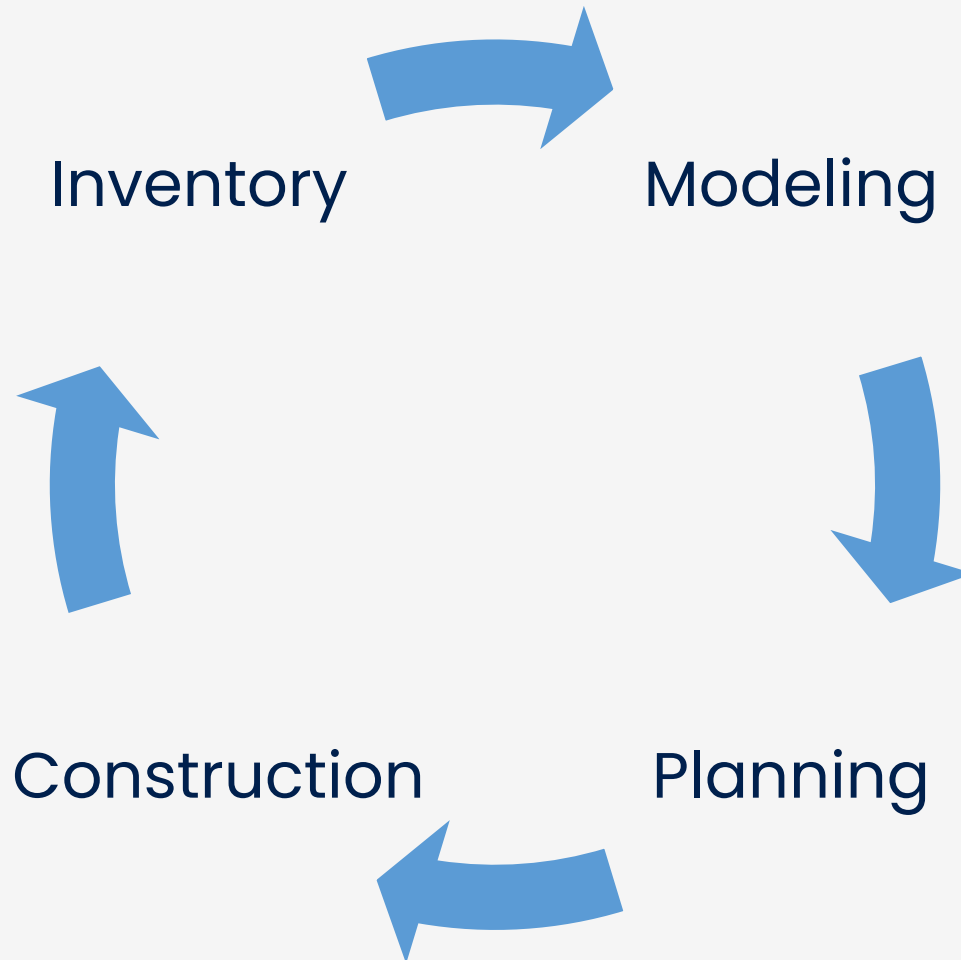
26 Variables Identified – weather events, volume, etc.

Collaborative formula development – Process in peer review

Will be managed by a SQL view – leaving inputs untouched



Pavement Marking Asset Lifecycle





Aaron Nash

aaron.nash@uconn.edu

Dr. Moataz Kilany

moataz.Kilany@uconn.edu

Jennifer Pawelzik

jennifer.pawelzik@uconn.edu

Gregory Ciparelli

gregory.ciparelli@ct.gov

Thank You!